

Tutorial 11

Matrix representations and eigenspaces

1. For each of the following linear transformations $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ find a 2×2 matrix such that for each $\mathbf{x} \in \mathbb{R}^2$, $T(\mathbf{x}) = A\mathbf{x}$.
 - (a) $T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} -x_1 \\ x_2 \end{pmatrix}$
 - (b) $T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_2 \\ -x_1 \end{pmatrix}$
2. In this question we go back to two questions in Tutorial 10. You may have to look at your work for those questions to answer this one.
 - (a) Find the matrix representing the linear transformation T in Question 2 of Tutorial 10 with respect to the basis B given in that question for $P_2(x)$ and the standard basis $C = \{1, x\}$ for $P_1(x)$.
 - (b) Find the matrix ${}_C[T]_B$ representing the linear transformation T in Question 3 of Tutorial 10 with respect to the standard basis B for $M_{2 \times 2}(\mathbb{R})$ and the standard basis C for P_2 .
3. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 2 \\ -2 & 0 & 3 \end{pmatrix}$.
 - (a) Find all the eigenvalues of A .
 - (b) For each eigenvalue λ found in (a), find the nullity of $A - \lambda I_3$.
 - (c) For each eigenvalue λ found in (a), find a basis for $NS(A - \lambda I_3)$.
 - (d) Are the basis vectors you found in (c) (there should be 3 in total) linearly independent? Give a reason for your answer.
 - (e) Is the 3×3 matrix P with these basis vectors as columns invertible? Give a reason for your answer. If it is invertible, find $P^{-1}AP$.

4. Let $0 < \theta < 2\pi$ be fixed and A_θ be the 2×2 matrix such that for every $\mathbf{x} \in \mathbb{R}^2$, $A_\theta \mathbf{x}$ is the vector obtained by rotating $\mathbf{x}$ anti-clockwise about the origin through θ radians.

(a) Show that there is only one value of θ for which A_θ has real eigenvalues. (You should be able to do this without doing any calculations.)

(b) Show that for all other values of θ A_θ has complex eigenvalues, and find them.

5. Let $A = \begin{pmatrix} 2 & 0 & 0 \\ -2 & 4 & 2 \\ 2 & -1 & 1 \end{pmatrix}$.

(a) Find all the eigenvalues of A .

(b) For each eigenvalue λ of A , find the corresponding eigenspace

$$W_\lambda = NS(\lambda I_3 - A)$$

and a basis for it.

(c) How many linearly independent eigenvectors does A have?

6. Let $B = \begin{pmatrix} 7 & 4 & -4 \\ 4 & -8 & -1 \\ -4 & -1 & -8 \end{pmatrix}$.

(a) You are told that 9 and -9 are eigenvalues of B . Find the other eigenvalue of B *without* using the characteristic equation of B .

(b) For each eigenvalue λ of B , find the eigenspace $W_\lambda = NS(\lambda I_3 - B)$ and a basis for it.

(c) Does $\mathbb{R}^3$ have a basis consisting of eigenvectors of B ? Give a reason for your answer.